

COMPUTER ARCHITECTURE – CSA1214

Assessment tool 1 – CO1 (Set 5)

1.8085 program:

8085 Microprocessor Simulator

File Edit Assembly Debug Help

Registers: A, B, C, D, E, H, L, PC, SP, PSW

Decimal-Hex Conversion: Decimal, Hex

I/O Ports: 0, Update Port Value

Memory: 0, Update Memory

Stack: 0000, 0001, 0002, 0003, 0004, 0005, 0006, 0007, 0008, 0009, 000A, 000B, 000C, 000D, 000E, 000F

Memory (Hex) Address Data

1F95	8085	0
1F96	8086	5
1F97	8087	190
1F98	8088	0
1F99	8089	0
1F9A	8090	0
1F9B	8091	0
1F9C	8092	0
1F9D	8093	0
1F9E	8094	0
1F9F	8095	0
1FA0	8096	0
1FA1	8097	0
1FA2	8098	0

Line No Assembly Message

0	Program assembled successfully
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Simulation Idle

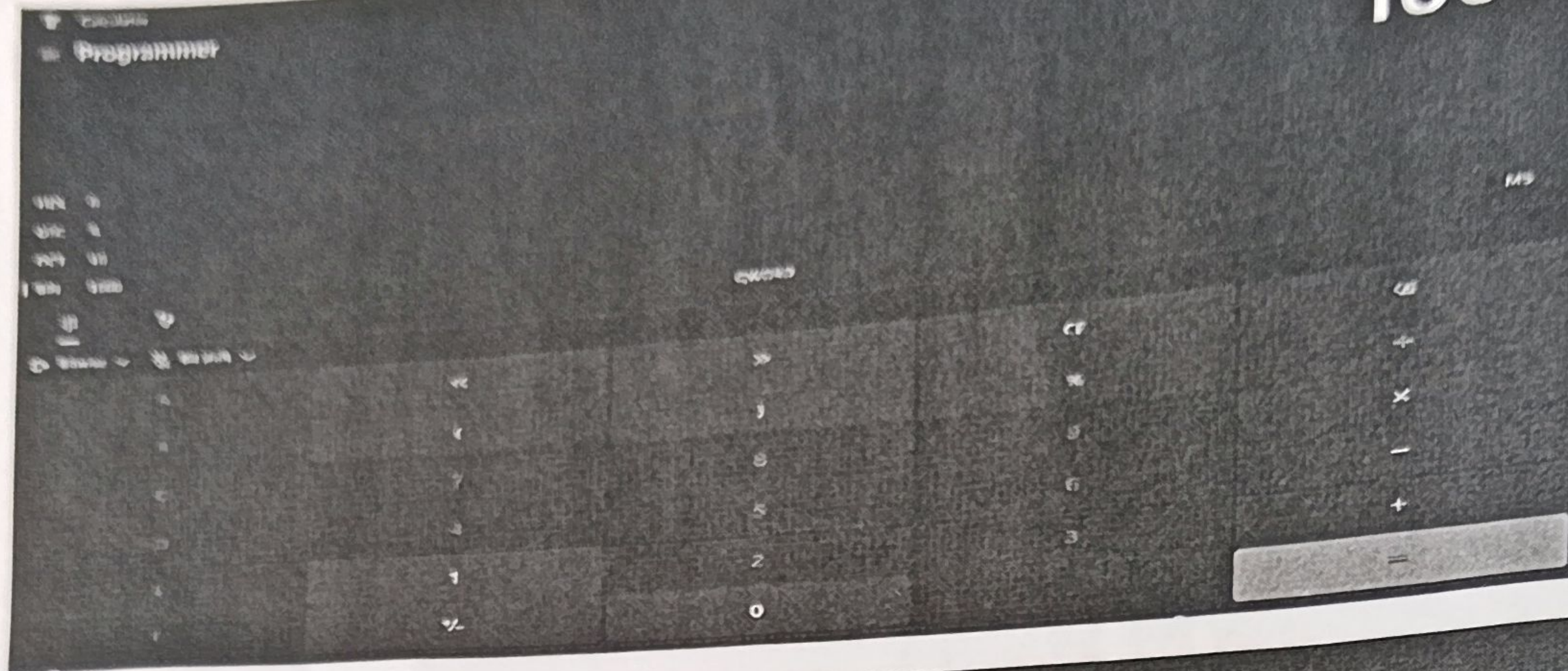
ENG IN 10:38 15-07-2025

EXPLANATION:

The program uses data transfer (MOV), arithmetic (ADI), and branch (JMP) instructions. The CPU executes them using the Fetch-Decode-Execute cycle.

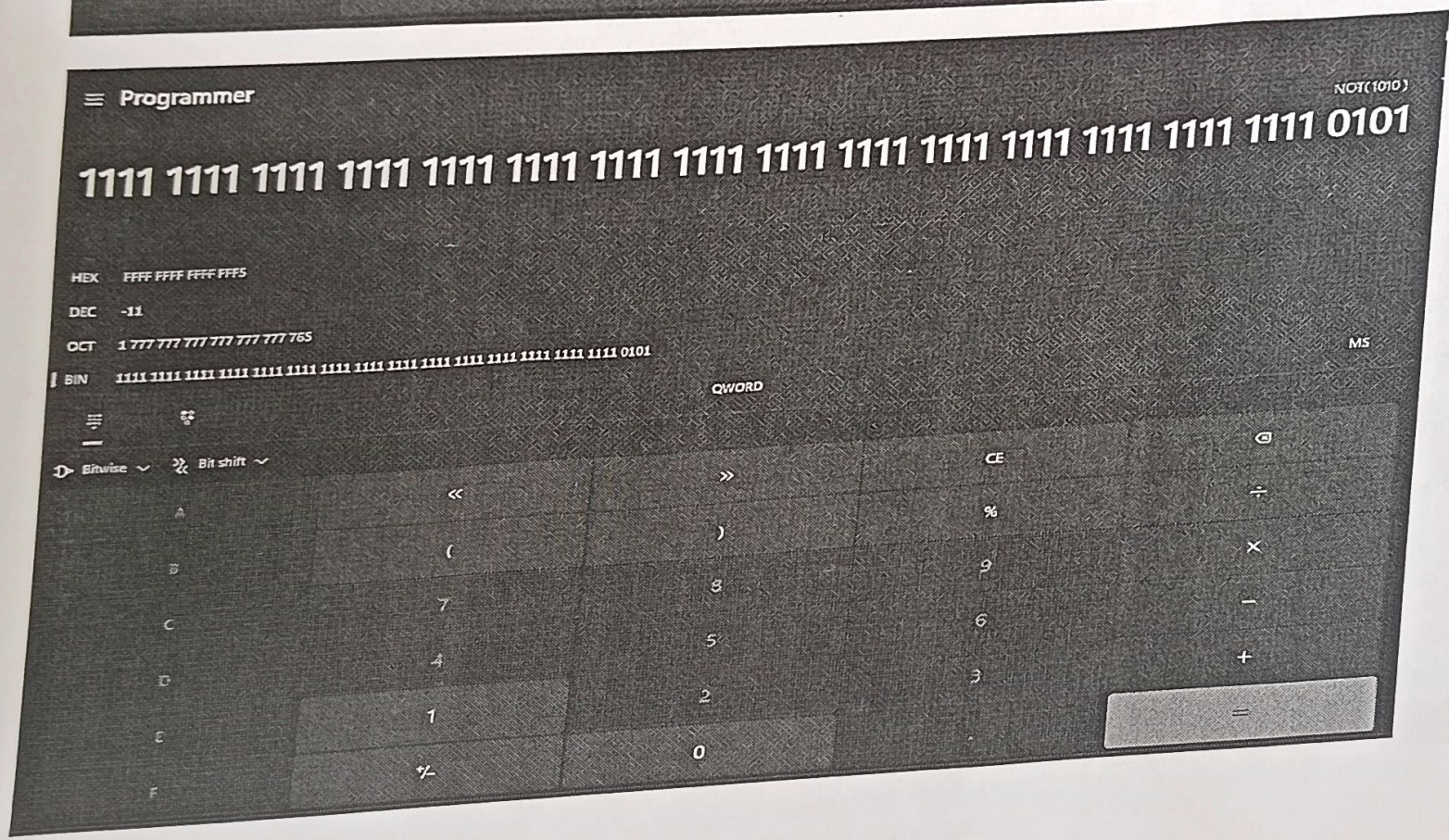
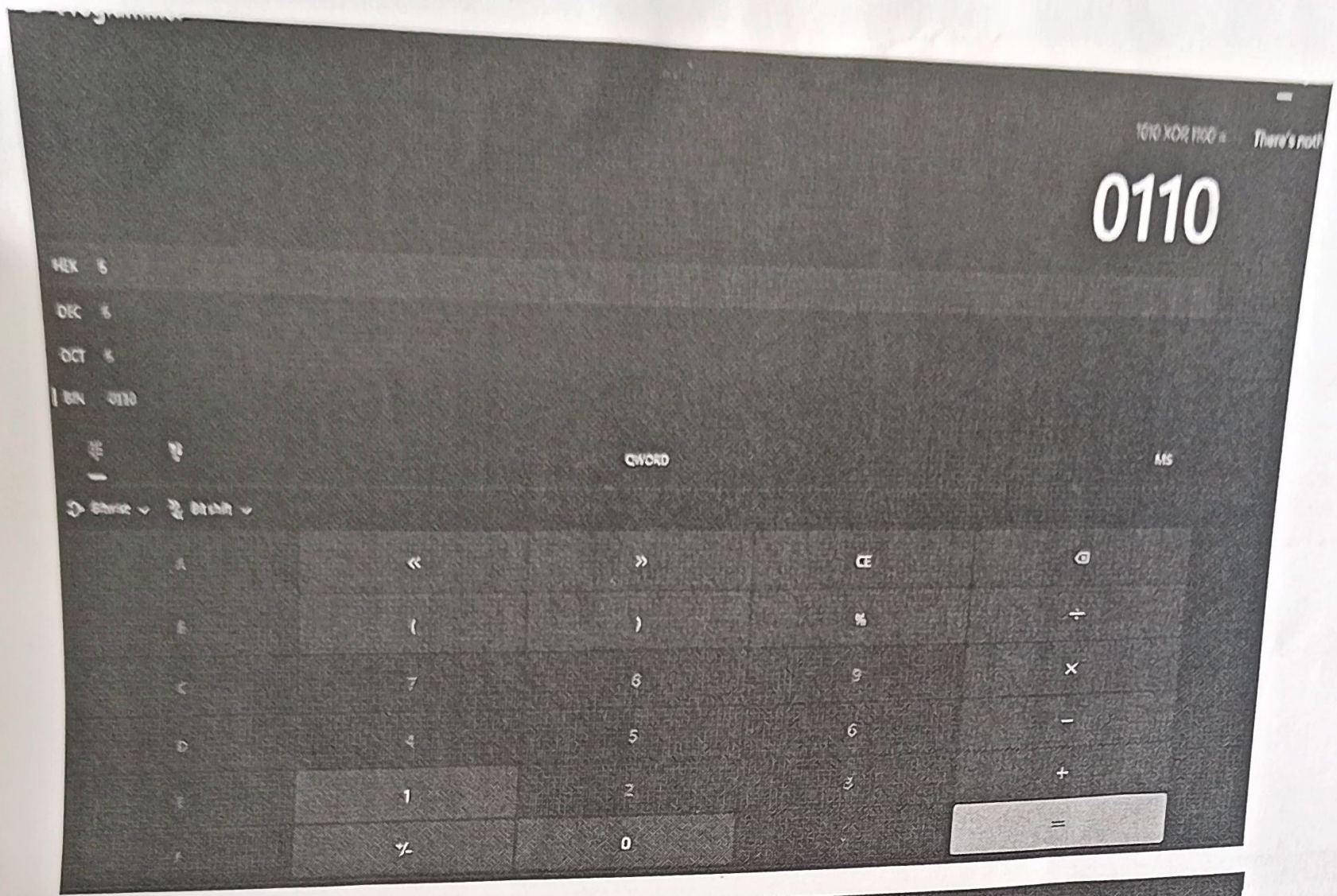
2. Windows calculator

1000



1110

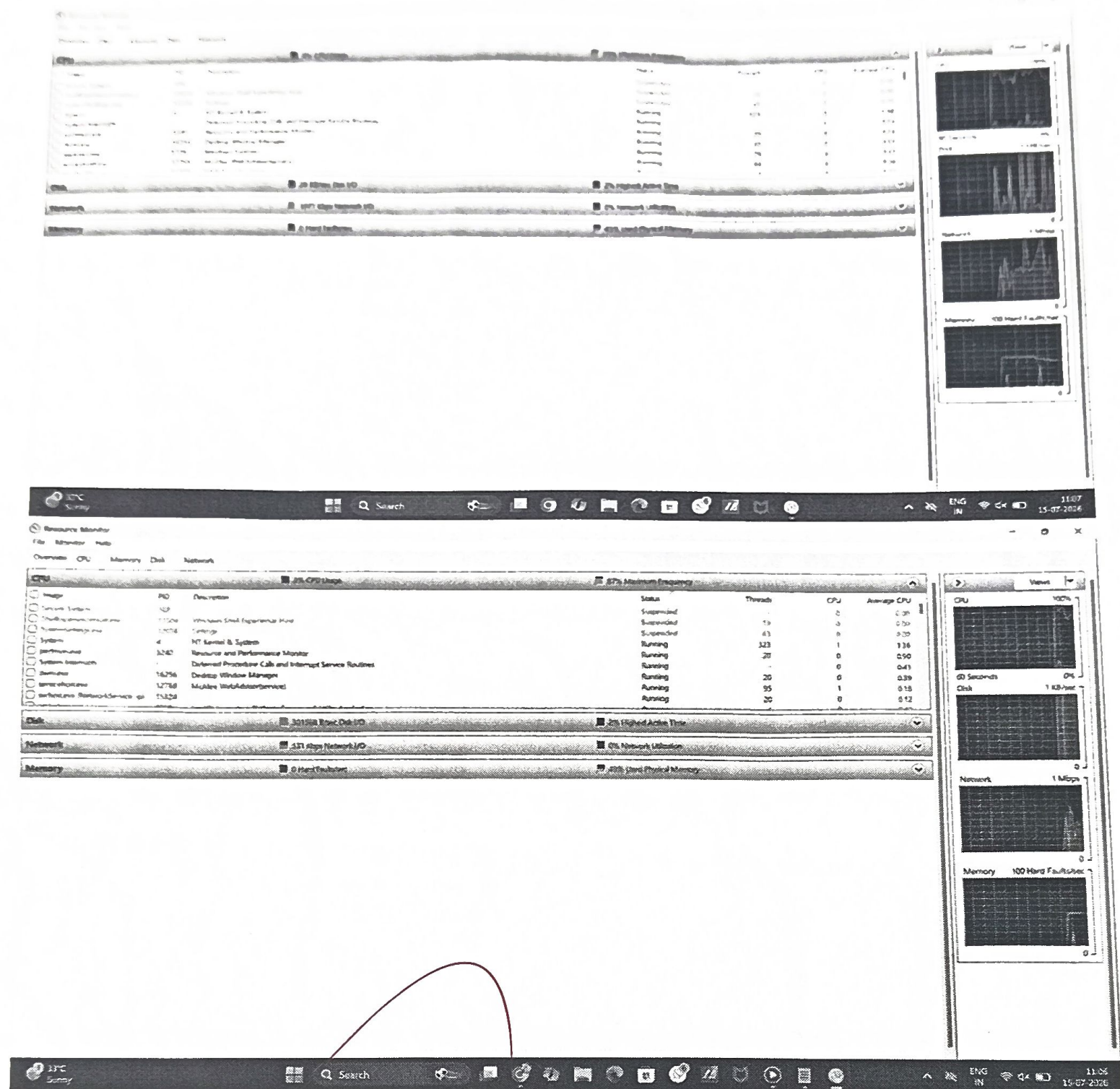




EXPLANATION:

Bitwise operations (AND, OR, XOR, NOT) are used to manipulate binary data and control hardware efficiently.

3. FUNCTIONAL UNITS



EXPLANATION:

CPU executes the programs.

RAM stores the running applications.

I/O devices transfer data between the computer and devices.